

PRIVACY-BY-DESIGN: ABSTRACTED METRICS, DATA SEPARATION, OPTIONAL ANONYMIZATION, AND SAFE CORRELATION OUTPUTS (NON-LIMITING)

A. Design Objective (Non-Limiting)

In various embodiments, the system is configured to maximize user privacy while enabling meaningful physiologic insights. Non-limiting objectives include minimizing storage of raw data, abstracting outputs into indices and confidence-weighted summaries, and limiting cross-device correlation to consent-gated, quality-verified windows.

B. Abstracted Outputs and On-Device Feature Extraction (Non-Limiting)

In various embodiments, raw sensor signals are converted into feature representations and abstracted outputs. Non-limiting approaches include: 1. transforming raw waveforms into indices, zones, ranges, distributions, and trend curves; 2. generating confidence indicators based on quality indices and validity gating; 3. representing results without explicit anatomical imagery and without explicit anatomical labeling; and 4. performing feature extraction on-node and retaining only derived features and summaries. In certain embodiments, the device stores only downsampled or aggregated representations unless high-resolution storage is explicitly enabled.

C. Data Separation and Compartmentalization (Non-Limiting)

In various embodiments, the system separates data into compartments to reduce linkage risk, including: 1. separating node identities such that each node maintains an independent identifier; 2. storing correlation mappings only under explicit consent gating; 3. separating personal identifiers from sensor data using tokenization; and 4. restricting cross-node join operations to environments with appropriate authorization. In certain embodiments, the system supports unlinking of nodes to remove correlation mappings while retaining standalone summaries.

D. Optional Anonymization and Aggregation for Population Models (Non-Limiting)

In optional embodiments, the system supports anonymized or pseudonymized aggregation for population-level modeling. Non-limiting approaches include: 1. stripping direct identifiers and retaining only coarse metadata; 2. aggregating features over time windows prior to upload; 3. applying noise or quantization to reduce re-identification risk; and 4. requiring explicit opt-in consent before any aggregation is performed. In certain embodiments, default behavior is opt-out for any population aggregation.

E. Consent-Gated Correlation and Safe Scoring (Non-Limiting)

In various embodiments, cross-device correlation outputs are generated only when: 1. explicit user consent is present for the involved nodes; 2. selected windows pass quality thresholds (e.g., coupling integrity and motion stability); and 3. correlation uses privacy-preserving features rather than raw waveforms. Non-limiting correlation outputs include alignment scores, stability scores, and confidence indicators. In certain embodiments, correlation outputs are constrained to be non-reconstructive such that raw waveforms cannot be recovered from the reported features.

F. User Controls and Transparency (Non-Limiting)

In various embodiments, a user is provided controls including: 1. enabling/disabling storage of raw data; 2. enabling/disabling synchronization; 3. enabling/disabling cross-node correlation; 4. viewing and exporting audit logs; and 5. initiating secure wipe procedures. In certain embodiments, the system provides user-readable explanations of what categories of data are stored, for how long, and under what consent state.

G. Variations

Any embodiment described herein may use different abstraction schemes, anonymization mechanisms, compartmentalization models, and consent flows while maintaining the functional intent of privacy-preserving metrics, minimized raw data retention, and consent-gated correlation outputs.